



KIRLOSKAR CHILLERS PRIVATE LIMITED

A Kirloskar Group Company

Enriching Lives

ENGINEERING SPECIFICATIONS

Document Name: Conso_XXX_2X Modbus List		
Document No: ESP-17-18-112	Revision: R02	Date: 19/03/2018
Prepared By: HPK	Approved By: SSN	Sheet: 1 of 7

KAS_XXX_2X Model				
Descriptions	UOM	Type	Read/Write	Modbus Add
Evaporator Water In Temperature	°C	ANALOG	Read	40002 (Holding Register)
Evaporator Water Out Temperature	°C	ANALOG	Read	40003 (Holding Register)
LWT Setpoint	°C	ANALOG	Read/Write	40151 (Holding Register)
Compressor - 1				
Discharge Temperature	°C	ANALOG	Read	40005 (Holding Register)
Suction Temperature	°C	ANALOG	Read	40006 (Holding Register)
Liquid Line Temperature	°C	ANALOG	Read	40007 (Holding Register)
Ambient Temperature	°C	ANALOG	Read	40008 (Holding Register)
Suction Pressure	Bar	ANALOG	Read	40010 (Holding Register)
Discharge Pressure	Bar	ANALOG	Read	40011 (Holding Register)
Oil Pressure	Bar	ANALOG	Read	40012 (Holding Register)
Motor Current	Amp	ANALOG	Read	40013 (Holding Register)
Suction Superheat	°C	ANALOG	Read	40014 (Holding Register)
Suction Saturation Temperature	°C	ANALOG	Read	40015 (Holding Register)
Percentage RLA	%	ANALOG	Read	40021 (Holding Register)
Evaporator Approach	-	ANALOG	Read	40022 (Holding Register)
Run Hours- High*	Hrs.	INTEGER	Read	45091 (Holding Register)
Run Hours- Low*	Hrs.	INTEGER	Read	45090 (Holding Register)



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Compressor – 2				
Discharge Temperature	°C	ANALOG	Read	40035 (Holding Register)
Suction Temperature	°C	ANALOG	Read	40036 (Holding Register)
Liquid Line Temperature	°C	ANALOG	Read	40037 (Holding Register)
Ambient Temperature	°C	ANALOG	Read	40038 (Holding Register)
Suction Pressure	Bar	ANALOG	Read	40040 (Holding Register)
Discharge Pressure	Bar	ANALOG	Read	40041 (Holding Register)
Oil Pressure	Bar	ANALOG	Read	40042 (Holding Register)
Motor Current	Amp	ANALOG	Read	40043 (Holding Register)
Suction Superheat	°C	ANALOG	Read	40044 (Holding Register)
Suction Saturation Temperature	°C	ANALOG	Read	40045 (Holding Register)
Percentage RLA	%	ANALOG	Read	40051 (Holding Register)
Evaporator Approach	-	ANALOG	Read	40052 (Holding Register)
Run Hours- High*	Hrs.	INTEGER	Read	45089 (Holding Register)
Run Hours- Low*	Hrs.	INTEGER	Read	45088 (Holding Register)



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KWS_XXX_2X Model				
Descriptions	UOM	Type	Read/Write	Modbus Add
Evaporator Water In Temperature	°C	ANALOG	Read	40002 (Holding Register)
Evaporator Water Out Temperature	°C	ANALOG	Read	40003 (Holding Register)
LWT Setpoint	°C	ANALOG	Read/Write	40151 (Holding Register)
Compressor - 1				
Discharge Temperature	°C	ANALOG	Read	40005 (Holding Register)
Suction Temperature	°C	ANALOG	Read	40006 (Holding Register)
Liquid Line Temperature	°C	ANALOG	Read	40007 (Holding Register)
Condenser IN Temperature	°C	ANALOG	Read	40008 (Holding Register)
Condenser Out Temperature	°C	ANALOG	Read	40009 (Holding Register)
Suction Pressure	Bar	ANALOG	Read	40010 (Holding Register)
Discharge Pressure	Bar	ANALOG	Read	40011 (Holding Register)
Oil Feed Pressure	Bar	ANALOG	Read	40012 (Holding Register)
Motor Current	Amp	ANALOG	Read	40013 (Holding Register)
Suction Superheat	°C	ANALOG	Read	40014 (Holding Register)
Suction Saturation Temperature	°C	ANALOG	Read	40015 (Holding Register)
Percentage RLA	%	ANALOG	Read	40021 (Holding Register)
Evaporator Approach	-	ANALOG	Read	40022 (Holding Register)
Condenser Approach	-	ANALOG	Read	40032 (Holding Register)
Run Hours- High*	Hrs.	INTEGER	Read	45091 (Holding Register)
Run Hours- Low*	Hrs.	INTEGER	Read	45090 (Holding Register)



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Compressor – 2				
Discharge Temperature	°C	ANALOG	Read	40035 (Holding Register)
Suction Temperature	°C	ANALOG	Read	40036 (Holding Register)
Liquid Line Temperature	°C	ANALOG	Read	40037 (Holding Register)
Condenser IN Temperature	°C	ANALOG	Read	40038 (Holding Register)
Condenser Out Temperature	°C	ANALOG	Read	40039 (Holding Register)
Suction Pressure	Bar	ANALOG	Read	40040 (Holding Register)
Discharge Pressure	Bar	ANALOG	Read	40041 (Holding Register)
Oil Pressure	Bar	ANALOG	Read	40042 (Holding Register)
Motor Current	Amp	ANALOG	Read	40043 (Holding Register)
Suction Superheat	°C	ANALOG	Read	40044 (Holding Register)
Suction Saturation Temperature	°C	ANALOG	Read	40045 (Holding Register)
Percentage RLA	%	ANALOG	Read	40051 (Holding Register)
Evaporator Approach	-	ANALOG	Read	40052 (Holding Register)
Condenser Approach	-	ANALOG	Read	40033 (Holding Register)
Run Hours- High*	Hrs.	INTEGER	Read	45089 (Holding Register)
Run Hours- Low*	Hrs.	INTEGER	Read	45088 (Holding Register)



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KWE_XXX_2X Model				
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Discharge Temperature	°C	ANALOG	Read	40005 (Holding Register)
Suction Temperature	°C	ANALOG	Read	40006 (Holding Register)
Liquid Line Temperature	°C	ANALOG	Read	40007 (Holding Register)
Condenser IN Temperature	°C	ANALOG	Read	40008 (Holding Register)
Condenser Out Temperature	°C	ANALOG	Read	40009 (Holding Register)
Suction Pressure	Bar	ANALOG	Read	40010 (Holding Register)
Discharge Pressure	Bar	ANALOG	Read	40011 (Holding Register)
Oil Pressure	Bar	ANALOG	Read	40012 (Holding Register)
Motor Current	Amp	ANALOG	Read	40013 (Holding Register)
Suction Superheat	°C	ANALOG	Read	40014 (Holding Register)
Suction Saturation Temperature	°C	ANALOG	Read	40015 (Holding Register)
Percentage RLA	%	ANALOG	Read	40021 (Holding Register)
Evaporator Approach	-	ANALOG	Read	40022 (Holding Register)
Condenser Approach	-	ANALOG	Read	40032 (Holding Register)
Run Hours- High*	Hrs.	INTEGER	Read	45091 (Holding Register)
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Compressor – 2				
Oil Temperature	°C	ANALOG	Read	40034 (Holding Register)
Discharge Temperature	°C	ANALOG	Read	40035 (Holding Register)
Suction Temperature	°C	ANALOG	Read	40036 (Holding Register)
Liquid Line Temperature	°C	ANALOG	Read	40037 (Holding Register)
Condenser IN Temperature	°C	ANALOG	Read	40038 (Holding Register)
Condenser Out Temperature	°C	ANALOG	Read	40039 (Holding Register)
Suction Pressure	Bar	ANALOG	Read	40040 (Holding Register)
Discharge Pressure	Bar	ANALOG	Read	40041 (Holding Register)
Oil Pressure	Bar	ANALOG	Read	40042 (Holding Register)
Motor Current	Amp	ANALOG	Read	40043 (Holding Register)
Suction Superheat	°C	ANALOG	Read	40044 (Holding Register)
Suction Saturation Temperature	°C	ANALOG	Read	40045 (Holding Register)
Percentage RLA	%	ANALOG	Read	40051 (Holding Register)
Evaporator Approach	-	ANALOG	Read	40052 (Holding Register)
Condenser Approach	-	ANALOG	Read	40033 (Holding Register)
Run Hours- High*	Hrs.	INTEGER	Read	45089 (Holding Register)
Run Hours- Low*	Hrs.	INTEGER	Read	45088 (Holding Register)

***Note: Total run hours will be combination of two registers**
Total Run hours = Run Hours High x 1000 + Run Hours Low



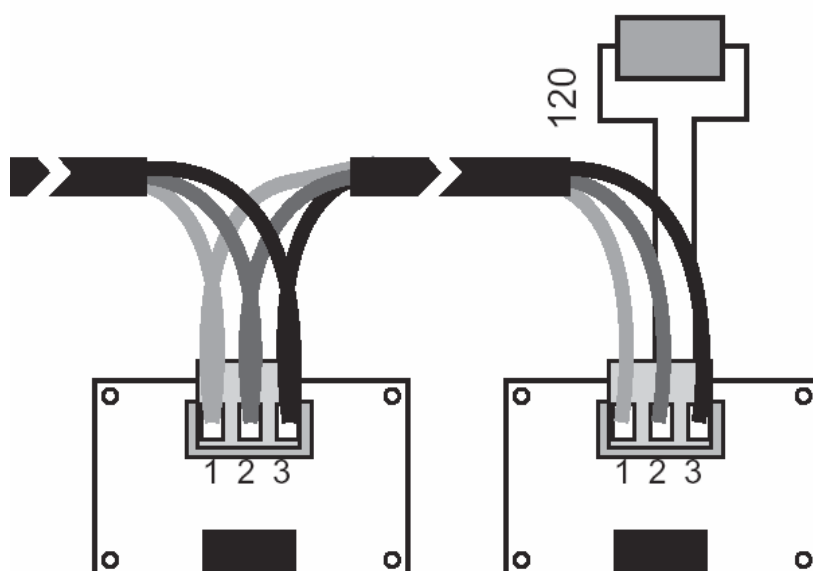
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Protocol Details	
Parity	None
Baud rate	19200
Protocol	Modbus/Modbus Extended
Start bit	1
Stop bit	1
No of bit	8
Communication Interface	RS-485
Interface Connection	2 Wire

The connection with the RS485 network is carried out by means of the plug-in terminal connector on the card. Pin-wiring of the connector is stamped on the card. If the card is placed in the last position of the supervision serial line, pins 2 and 3, you must connect a 120 Ω - 1/4 W end line resistor, as shown.



pin	significato <i>meaning</i>
1	GND
2	RX+/TX+
3	RX-/TX-